

1
Max σ will be hoop stress

$$\sigma_{\theta} = -\sigma^* (1 - 3\eta^2)$$

$$\sigma^* = \frac{KE\Delta T}{4(1-\nu)}$$

$$\Delta T = T_o - T_s$$

$$\sigma^* = \frac{10E-6 \cdot 200 \cdot \Delta T}{4(1-0.35)}$$

Need ΔT

$$\Delta T = \frac{LHR}{4\pi K} = \frac{300}{4 \cdot \pi \cdot 0.025} = 955 \text{ K}$$

$$\sigma^* = \frac{10E-6 \cdot 200 \cdot 955 \text{ K}}{4(1-0.35)} \rightarrow \boxed{734 \text{ MPa}}$$

$$\sigma_{\theta,f} = 175 \text{ MPa}$$

$$175 \text{ MPa} = -734 \text{ MPa} (1 - 3\eta^2)$$

$$\frac{175}{734} = 1 - 3\eta^2$$

$$\eta = \sqrt{\left(1 - \frac{175}{734}\right)} = 0.873$$

∴ CRACK will extend 0.873% of way into fuel

$$\frac{2}{p} = 15 \text{ MPa} \quad \bar{R}_e = 0.55 \text{ cm} \quad t_o = 0.05 \text{ cm}$$

$$a) \quad \sigma_\theta = \frac{pR}{\delta} = \frac{15 \cdot (0.55)}{0.05} = \boxed{165 \text{ MPa}}$$

$$\sigma_z = \frac{pR}{2\delta} = \boxed{82.5 \text{ MPa}}$$

$$\sigma_r = -\frac{p}{2} = \boxed{-7.5 \text{ MPa}}$$

$$R_i = 0.525$$

$$R_o = 0.575$$

$$\frac{R_o}{R_i} = 1.095$$

$$b) \quad \sigma_\theta = p \left[\frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} \right]$$

$$\boxed{r = R_i}$$

$$\sigma_\theta = 15 \left[\frac{(1.095)^2 + 1}{(1.095)^2 - 1} \right] = \boxed{165 \text{ MPa}}$$

$$\sigma_z = \frac{15}{(1.095)^2 - 1} = \boxed{75.4 \text{ MPa}}$$

$$\sigma_r = 15 \left[\frac{(1.095)^2 - 1}{(1.095)^2 - 1} \right] = \boxed{-15 \text{ MPa}}$$

First, Need Refuse T profile

$$T_{co} - T_{ci} = \frac{LHR}{2\pi R_F} \frac{t_c}{K_c} = \frac{250}{2\pi \cdot .48} \cdot \frac{0.08}{0.15} = 44.2 K$$

$$T_s - T_i = \frac{LHR}{2\pi R_F} \frac{t_g}{K_g} = \frac{250}{2\pi \cdot .48} \cdot \frac{0.008}{0.003} = 221 K$$

$$T_o - T_g = \frac{LHR}{4\pi R_F} = \frac{250}{4\pi \cdot 0.03} = 663 K$$

$$\Delta \sigma_g = \Delta R_c - \Delta R_F = K R_c \Delta T_c - K R_F \Delta T_F$$

$$K = 10 \cdot 10^{-6}$$

$$T_{co} = 550 K$$

$$T_c = \frac{T_{ci} + T_{co}}{2}$$

$$\frac{550 + (550 + 44.2)}{2}$$

$$T_c = 572.1$$

$$R_c = \frac{0.48}{0.48 + \left(\frac{0.18}{2} + 0.48\right)} = 0.6$$

$$T_F \approx \frac{T_s + T_o}{2} = \frac{815 K + 1478}{2} = 1147 K$$

$$\Delta t_g = \Delta R_c - \Delta R_F = K R_c \Delta T_c - K R_F \Delta T_F$$

$$\Delta t_g = 10^{-6} \cdot 0.6 \cdot (572.1 - 300) - 10^{-6} \cdot 0.48 \cdot (1147 - 300)$$

$$\Delta t_g = 0.001632 - 0.00056918$$

$$\Delta t_g = 4.05 \cdot 10^{-3} \text{ cm}$$

4

$$R_i = 0.55 \quad t_c = 0.06 \quad \text{Reynolds}^4$$

$$\sigma_o = ? \quad \sigma_r = ? \quad r = R_i$$

$$\sigma_r = \frac{\Delta T}{2} \frac{KR}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\Delta T = T_{\infty} - T_{ci} = 50 \text{ K}$$

$$\sigma_r = \frac{50}{2} \cdot \frac{15 \cdot 10^{-6} \cdot 100 \cdot 10^6}{1 - 0.34} \left(\frac{r}{R_i} - 1 \right)$$

$$\therefore \sigma_r = 0, \text{ since } r = R_i$$

$$\sigma_o = \frac{\Delta T}{2} \frac{KR}{1-\nu} \left(1 - 2 \frac{R_i}{r} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_o = \frac{50}{2} \frac{15 \cdot 10^{-6} \cdot 100 \cdot 10^6}{1 - 0.34} \left(1 - 2 \frac{0.55}{0.06} \left(\frac{0.55}{0.55 - 0.06} - 1 \right) \right)$$

$$\sigma_o = 7.07 \text{ MPa}$$

5

$$\dot{F} = 6 \cdot 10^{13} \quad \Delta p_o = 0.015 \quad B_o = 5 \frac{\text{mWD}}{\text{kgU}} \quad p = 10.47$$

$$t = 10 \text{ days} \quad \hookrightarrow B_o = \frac{5}{0.50} = 0.00524 \text{ FIMA}$$

$$E_o = \Delta p_o \left(\exp \left(\frac{B h(0.01)}{c_o B_o} \right) - 1 \right)$$

Need $B \rightarrow B = \frac{\dot{F} t}{N_u}$

$$N_u = 10.47 \frac{\text{g}}{\text{kg}} \frac{1 \text{ mol}}{270 \text{ g}} \frac{6.022 \cdot 10^{23}}{1 \text{ mol}}$$

$$N_u = 2.45 \cdot 10^{22}$$

$$B = \frac{6 \cdot 10^{13} \cdot (10 \text{ days} \cdot 24 \cdot 3600)}{2.45 \cdot 10^{22}} = 0.602116$$

$$E_o = 0.015 \cdot \left(\exp \left(\frac{0.002116 \cdot h(0.01)}{1 \cdot 0.00524} \right) - 1 \right)$$

$$E_o = -0.6126$$

6. elasticity $\rightarrow$ Ability to resist force & return to original shape
Plasticity $\rightarrow$ After force is removed, non-reversible distortion/change in shape. Permanent deformation when force is removed
size when force is removed

7

Plasticity changes microstructure, usually increasing strength (σ_y)
Caused by stress/strain, but due to shifts in crystal lattice which "aligns" microstructure in more optimal way than before stress was present

8

Model temps in fuel, model temp in cladding, provide gap heat transport estimate. All three solve the fuel thermo/mech problem

9

When sintering is done, particles are in different positions/orientations relative to one another, so grain boundaries form where things don't "line-up"

10

Fuel pellets are not perfectly dense (see answer #a), and the microstructure impacts the material's thermal performance. This is critical in understanding how heat moves through fuel & cladding. For thermal conductivity, a microstructure-based model allows for a more accurate model which doesn't depend on burn-up, for example. Allows for less reliance on empirical models.

11

microstructure $\rightarrow$ structure of material grains, has impact on thermal performance. Can change ductility, temp behavior, etc.

Processing $\rightarrow$ ex. Quenching $\rightarrow$ heating to temp for some time, then rapidly cooled
Causes phase trans. w/out gradual stress removal.
similar to ϵ/σ curves, increases σ_y

12

~~Diffusion~~ ~~At boundaries~~ At boundaries, intergranular diffusion is more significant than intragranular. The grain is more ideal toward the center.

13

P_d drives $\rightarrow$ reduction in GB energy

P_r resists $\rightarrow$ caused by mobility inhibitors like pores, defects, etc.

14

Densities due to fuel not being 100% dense due to sintering process. Gaps in microstructure are "collapsed" as fuel continues to heat in operation. At micro-scale, occurs due to energy change associated with shrinking surface Area / free E.